Applied Advanced Optimisation iRAT 8

Zhao, Amor (CID: 02019680)

March 6, 2025

Question 1

Provide the barrier function. Explain how the barrier function is used to approximate the problem with the indicator function I_-. Explain what happens when one of the inequality constraints approaches zero.

$$\phi(x) = -\sum_{i=1}^{m} \log(-f_i(x))$$

When an inequality constraint approaches zero, the corresponding term in barrier function goes to infinity. The penalty for violating the constraint is thus very large, and prevents it from happening in our optimisation problem, making sure the solution is feasible.

Question 2

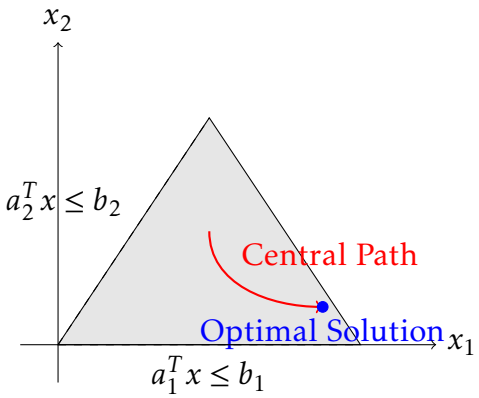
Provide the centrality condition. Consider an LP in inequality form and the hyperspace defined by its inequality constraints. For such a problem, sketch an example of centrality path and explain it.

Centrality condition makes sure that the points are strictly feasible and there exists $\hat{v} \in \mathbb{R}^p$ such that

$$0 = t\nabla f_0(x^*(t)) + \nabla\phi(x^*(t)) + A^\top \hat{v}$$

For an LP in inequality form, $f_0(x) = c^\top x$, $f_1(x) = Ax - b$, the centrality condition can be written as:

$$tc + \nabla\phi(x^*(t)) = 0.$$



Question 3

Give the modified complementary condition associated to the centrality condition and explain it as a function of t.

The modified complementary condition associated with the centrality condition is as follows:

$$-\lambda_i f_i(x) = \frac{1}{t}, \quad i = 1, \ldots, m.$$

As t increases, the term $\frac{1}{t}$ forces the complementary slackness to become smaller, and for large t, the solution $x^*(t)$ approaches the optimal point, the solution would satisfy the KKT conditions more closely.

Question 4

Explain the differences between the two Phase I methods seen in the lecture.

When the problem is infeasible, the maximum infeasibility method minimizes the largest constraint violation, the solutions we get in this way may have many constraints slightly violated.

The sum of infeasibilities method minimizes the total violation, most constraints are exactly satisfied, while a few may have larger violations.

Question 5

Provide the duality gap associated to a convex optimisation problem with m generalised inequalities which are K_i-convex and have generalised logarithms with degree θ_i. Explain the role of θ_i in the computational complexity of the barrier method.

The duality gap for a convex optimization problem with m generalized inequalities, where each inequality is K_i-convex and has a generalized logarithm with degree θ_i:

$$\text{duality gap} = \frac{\sum_{i=1}^{m} \theta_i}{t}$$

The generalized logarithm with degree θ_i ensures that the barrier function used to enforce the K_i-convex constraints has a self-concordant property.

Feedback

Q1

Some did not explain how the barrier is used to approximate the indicator function (answer: paragraph above Figure 9.1), or did not mention the differentiability of the such a function.

Q2

Some students did not make the sketch (answer: for instance Figure 9.2). One student did not provide the centrality condition.

Q3

Some students did not explain the modified complementary condition as a function of t. Answer: above Figure 9.3 there is a list of bullet points with two points. The first point provides the answer to this question.

Q4

Some students stated the two optimisation problems, but did not really explain the difference between phase I and phase II. Answer: the paragraph above Figure 9.7

Q5

A few students forgot to replace m with $\bar{m}$ or gave an incorrect bound. A few did not explain the role of θ_i. One student skipped this question.